A Vision-Grounded Per-Image Phenotype Dataset Across 239 Ethnic Groups: Construction, Caveats, and Release

Jason Jacoby AgaveIS admin@agaveis.com

DOI: 10.5281/zenodo.20075617 **Dataset:** https://huggingface.co/datasets/EthnicErotic/phenotype-catalog **Code:** https://github.com/Agaveis/phenotype-catalog-pipeline **License:** CC BY 4.0 **Version:** 2.0 (2026-05-07)

Abstract

We release **PhenotypeCatalog v2**, a public dataset of 5,668 per-image phenotype observations across 239 ethnic groups, constructed entirely from public-domain Wikipedia photographs of notable people. Each observation is the structured output of a vision language model (Anthropic Claude Sonnet 4.6 via AWS Bedrock) over 14 phenotype fields, including Fitzpatrick skin type, hair color/texture/pattern, eye color/shape (with epicanthic-fold detection), facial features, build, image quality, and a model-reported confidence score. The dataset accompanies a 484-row ethnographic catalog with synthesized phenotype profiles (484/484 groups) and aggregated observed-distribution summaries (209/484 groups). Source attribution is preserved at the row level via Wikipedia URLs, enabling per-image license verification and audit. We document explicit limitations — most importantly, that "Wikipedia notable people" is a non-population-representative sample biased toward male and public-life subjects — and position the dataset as a complement to existing fairness benchmarks (FairFace, BUPT-Balancedface) rather than a replacement. Intended uses include bias auditing of vision systems, Fitzpatrick-balanced training subsets, and anthropological reference. The dataset is **not** appropriate for individual classification.

---

1. Introduction

Demographic bias auditing of computer-vision systems requires phenotype-balanced evaluation data, but the available datasets cluster at coarse race buckets — typically four to seven categories — rather than finer ethnographic grain. The widely-used FairFace dataset (Karkkainen & Joo, 2021) provides seven race buckets across roughly 100,000 images; BUPT-Balancedface (Wang et al., 2020) similarly aggregates to four. These resources are valuable but blind to within-bucket variance: a single "Asian" or "African" label compresses real and meaningful phenotype distinctions, which limits the diagnostic resolution of any audit built on them.

Constructing a finer-grained dataset directly is expensive: crowdworker annotation introduces its own biases (Buolamwini & Gebru, 2018, document this extensively for gender), and most candidate image sources lack the per-image license clarity required for redistribution. We address this with a different construction recipe:

1. **Source images from public-domain Wikipedia photographs** of notable people, sourced via each ethnic group's "List of {Ethnicity} people" article and its referenced biographies. Per-image license is verifiable through the original Wikipedia URL.
2. **Annotate via vision language model** (Anthropic Claude Sonnet 4.6) using a fixed structured prompt that produces 14 phenotype fields plus model-reported confidence and image quality.
3. **Aggregate per-group with deterministic SQL** (no LLM), surfacing sample size, source breakdown, Fitzpatrick distribution, hair/eye/epicanthic-fold distributions, and explicit sample-bias caveats.

The resulting dataset (5,668 image-level rows + 209 per-group aggregations + 484 group-level metadata rows) is released under CC BY 4.0 alongside the source pipeline as open code.

We make no claim that the dataset is population-representative. The Wikipedia "notable people" sampling frame is the dominant source of bias, and we discuss it explicitly in §6. Our position is that surfacing this bias through structured per-image rows with explicit source URLs and confidence scores is more useful than burying it inside aggregated coarse-race labels.

Contributions

1. A publicly-released structured per-image phenotype dataset at unusual ethnographic grain (239 groups, mean ~24 images/group).
2. A reproducible pipeline (Wikipedia scrape → image URL discovery → vision-LLM annotation → aggregation) released as open code.
3. An empirically-grounded discussion of the failure modes specific to vision-LLM phenotype annotation: source skew, self-reported confidence as a soft signal, and the "available image quality" bucket bias.

---

2. Related Work

Fitzpatrick skin-type scale. Our skin-tone field uses the Fitzpatrick (1988) classification, originally developed for sun-reactivity assessment but now widely adopted as the de-facto skin-tone vocabulary in computer-vision fairness work. We use the I-VI scale and accept ranges (e.g., "Type II-III") as valid responses where the model judges a single bin to be over-precise.

Fairness benchmarks for face attribute systems. Buolamwini & Gebru (2018, *Gender Shades*) demonstrated that commercial gender classification systems exhibit substantial accuracy disparity across phenotype subgroups, motivating subsequent dataset releases. FairFace (Karkkainen & Joo, 2021) provides ~108K face images balanced across seven race buckets, two genders, and nine age buckets. BUPT-Balancedface (Wang et al., 2020) similarly addresses race-imbalance for face recognition training. These resources are foundational to the present work; we view ours as complementary rather than competitive — coarser per-image label space, but finer ethnographic grain, with explicit source attribution.

Vision-LLM as annotator. Using a vision-capable language model as a structured annotator is a recent pattern. We do not claim novelty here; we document one full pipeline at scale, with the cost numbers, failure modes, and prompt details that are typically omitted from blog posts.

Dataset distribution platforms. The release uses HuggingFace Datasets (Lhoest et al., 2021), which provides a standardized loading interface and is indexed by Google Dataset Search.

---

3. Dataset Construction

The pipeline runs end-to-end from a SQL Server `ethniclist` source table to four output configurations (CSV + JSONL). Each stage is idempotent, source-attributed, and re-runnable.

3.1 Group-level catalog

The starting point is a curated 484-row catalog of ethnic groups, indexed by name with normalized columns for homeland, region, language, language ISO codes, country ISO codes, religion, and a Wikipedia URL where available. The taxonomy follows a standard continent → sub-region → ethnicity hierarchy (Americas, Europe, Africa, Asia, Oceania at the continent level; 23 sub-regions). This catalog predates the present work and is published as the `ethnicities` config of the dataset.

3.2 Notable-people scrape

For each group with a Wikipedia URL, we attempt to fetch the corresponding "List of {Ethnicity} people" article. When such an article exists, we extract names, short "known for" descriptors, birth/death years where surfaced, and per-person reference URLs. **291 of 484 groups** have at least one such article on Wikipedia; **193 do not** (typically small or obscure groups for which no list page has been authored). The scrape produced **13,094** people-rows total. This is published as the `notable_people` config.

3.3 Image URL discovery

For each scraped person, we follow the reference URL to that person's individual Wikipedia article and extract a representative image URL — preferring the article's lead `infobox` image, falling back to the first OpenGraph image. **6,243 of 13,094** persons (47.7%) have a discoverable Wikipedia image; the remaining ~52% have no infobox image (often pre-photography figures or low-prominence subjects with text-only articles).

We rate-limit the upload.wikimedia.org fetcher to 2 requests per second per source IP; tighter rates trigger HTTP 429 within batches.

3.4 Vision-LLM annotation

Each image with a discoverable URL is submitted to a vision language model with a fixed structured prompt asking for 14 phenotype fields plus a self-reported confidence score (0.0–1.0) and an image quality bucket (`high` / `medium` / `low` / `very_low`). The prompt explicitly instructs the model to:

- Use Fitzpatrick I–VI vocabulary for skin tone, accepting ranges where appropriate.
- Detect and report epicanthic-fold presence as a structured eye-shape sub-field.
- Treat the photograph as one observation of one person, not an ethnic stereotype prototype.
- Return `unknown` or empty for fields that cannot be assessed (e.g., build when only the head is visible).
- Surface obscurations (glasses, hat, makeup, partial-face) explicitly.

The model used is the Anthropic Claude Sonnet 4.6 inference profile on AWS Bedrock (`us.anthropic.claude-sonnet-4-6`). We chose Sonnet 4.6 over higher-tier alternatives because the task is short-context structured extraction; pilot runs showed no measurable accuracy lift from Opus-tier models on this prompt, and Sonnet's per-image cost is substantially lower.

Run statistics:

- **5,668** images successfully analyzed.
- **~575** images failed (load failures, model errors, parse failures); these are not in the released dataset.
- Concurrency: 4 simultaneous calls.
- Throughput: ~0.42 images/second (network-bound; the model call itself completes in ~1.5 seconds per image).
- **Total cost: $44.66 USD** (input tokens for image + prompt, output tokens for structured response, at Bedrock list pricing 2026-05-06).

The structured response is parsed and stored row-by-row; the original raw JSON is retained in the source database (column `ethnic_image_analysis.raw_json`) for audit but not included in the release.

3.5 Per-group aggregation

A deterministic SQL/JavaScript aggregator produces a per-group summary card from each group's image-observation rows. The aggregator is non-LLM and re-runnable. For each group with ≥3 image observations (209/484), it produces:

- Sample size and source breakdown.
- Quality split (high/medium/low/very_low percentages).
- Mean self-reported confidence.

- Fitzpatrick distribution (proportion of rows in each bin I–VI).
- Hair color distribution, hair texture distribution.
- Eye color distribution.
- Epicanthic-fold proportion (yes / no / partial).
- Conditional caveat lines:
    - **N < 10**: "small sample — interpret with caution"
    - **N < 25**: "modest sample"
    - **<30% high-quality**: "image quality is mixed"
    - **mean confidence < 0.55**: "low overall confidence"
    - **source breakdown 100% Wikipedia**: "Wikipedia notable people skews male and public-life — not population-representative"

These caveat lines are included as visible text in both the on-site rendering and in the dataset's `image_observed_distribution` column.

---

4. Schema

The release contains four configurations, each a separate `config_name` under HuggingFace's `datasets` API:

Config	Rows	Description
`ethnicities`	484	One row per ethnic group
`atlas`	~21	Phenotype reference categories (eyes, lips, nose, hair, skin, body)
`notable_people`	13,094	Wikipedia-sourced people, joinable via `ethnic_id`
`image_observations`	5,668	Per-image phenotype rows, joinable to `notable_people` via `example_id`

`image_observations` columns (the headline asset):

Column	Type	Notes
`id`, `example_id`, `ethnic_id`	int	Identifiers and FKs
`ethnic_name`, `person_name`	string	Denormalized for direct use
`image_url`, `reference_url`	string	Source attribution at row level
`skin_tone`	string	Fitzpatrick I–VI, ranges allowed (e.g., `"II-III"`)
`skin_undertone`	string	`cool` / `warm` / `neutral` / descriptive
`hair_color`	string	Free text (e.g., `"dark brown"`, `"black with grey"`)
`hair_texture`	string	`straight` / `wavy` / `curly` / `coily` / etc.
`hair_pattern`	string	Free text (longer descriptor when present)
`eye_color`	string	`brown` / `hazel` / `blue` / `green` / etc.
`eye_shape`	string	Includes epicanthic-fold subfield
`facial_features`	string	Free-text prose

`build`	string	Body build descriptor when visible
`visible_extent`	string	Which body parts are visible
`image_quality`	string	`high` / `medium` / `low` / `very_low`
`obscurations`	string	Glasses, hat, makeup, partial-face, etc.
`confidence`	float	0.0–1.0, model self-reported
`analysis_model`	string	Model identifier
`analyzed_at`	string	ISO-8601 timestamp

The `ethnicities` config gains four columns in v2: `phenotype_profile` (synthesized prose, 484/484), `image_observed_distribution` (aggregated text, 209/484), `image_observed_at` (timestamp), and `notable_people_count` (denormalized for convenience).

---

5. Coverage and Statistics

Metric	Value
Image observations	5,668
Distinct ethnic groups represented	239
Mean observations per represented group	23.7
Median observations per represented group	14
Largest group (by observations)	Punjabis (n=146)
Mean self-reported confidence	0.67
Quality split (high / medium / low / very_low)	42% / 39% / 16% / 3%
Epicanthic-fold annotated proportion	~22% (of observations where the field was assessable)

Sample size variance. The 239 groups with image observations have unequal coverage. 209 have ≥3 observations (the threshold for aggregated `image_observed_distribution`); 30 are between 1 and 2. Filtering to `n ≥ 25` per group leaves ~70 groups suitable for stand-alone per-group statistical claims. This variance is structural — Wikipedia article density tracks population size, English-language coverage, and historical prominence, not actual ethnic-group population.

Confidence and quality as filtering signals. We recommend that downstream consumers apply at minimum the filter `image_quality IN ('high', 'medium')` (excluding ~19% of rows) and consider `confidence > 0.55` as an additional threshold for high-precision use cases. The released dataset includes all rows; we leave filtering to the consumer.

---

6. Limitations and Biases

6.1 Wikipedia source skew (primary)

The dominant bias in this dataset is the construction frame: "people Wikipedia has a list-of-X-people article for, with a photograph in their individual article." This sample is:

- **Gender-skewed male.** Public-life prominence on Wikipedia tilts male in most categories (politicians, scientists, athletes, historical figures). We did not measure the per-group gender split, but inspection of the data suggests overall male skew of roughly 70–75%.
- **Biased toward public life.** Scientists, politicians, entertainers, athletes, and historical figures are over-represented relative to the general population.
- **English-language coverage biased.** Groups with stronger English-Wikipedia presence have more image observations; this is loosely correlated with Anglophone diaspora size, not native-population size.
- **Photographic-era biased.** Pre-photography figures have no infobox image, so groups whose Wikipedia presence skews historical have lower image-observation coverage.

The aggregator surfaces this caveat textually whenever the source breakdown is 100% Wikipedia — which is currently every row. Future releases that incorporate user-submitted images or a second public-domain source (see §10) will dilute this skew.

6.2 Model self-report

The `confidence` field is the model's own estimate of its annotation quality. It is **not** external validation. Empirical correlation between `confidence` and ground-truth correctness has not been measured at scale on this prompt; treat it as a soft signal for filtering, not as a calibrated probability.

6.3 Aggregate-only intent

The dataset is structured per-image but is **not** intended for individual-level inference. Each row is one observation of one photograph. We do not endorse:

- Identification of individuals from these rows.
- Surveillance applications keyed on the image URLs.
- Profiling or matching of individuals by phenotype.

Each `image_observations` row's source URL is a public-domain Wikipedia photograph that already exists publicly; the dataset does not increase any individual's discoverability beyond what a Wikipedia search already provides. We surface this norm explicitly in the dataset card under "Out-of-scope uses."

6.4 Anglo-centric phenotype vocabulary

The 14 fields use English anthropological vocabulary (`Fitzpatrick III` , `wavy` , `epicanthic fold`). Self-identifications and exonyms in other languages are not always captured. Researchers working with non-English-speaking populations should treat the field values as approximate translations of locally-meaningful categories.

6.5 Sample size

209 of 484 groups have ≥3 observations. 275 groups have fewer than 3 (or none). Smaller groups should not be inferred from individual-image rows alone; the `image_observed_distribution` column is suppressed below the threshold for exactly this reason.

---

7. Intended Uses

We describe four use cases the dataset is designed to support, and one it explicitly is not.

7.1 Bias auditing of vision systems

Per-group fairness analysis becomes tractable when you have ground-truth structured labels keyed by ethnic-group identifier. A vision system's per-group performance can be measured against the `image_observations` rows directly: pass each `image_url` through the system under test, compare its output (skin tone, age, gender, etc.) against the dataset's structured fields, and report per-group accuracy. The `ethnic_id` column is the join key.

7.2 Fitzpatrick-balanced training subsets

The `skin_tone` column allows construction of Fitzpatrick-balanced training/evaluation subsets at higher granularity than the four-bucket race labels typical in existing benchmarks. We recommend constructing such subsets *within* the dataset's quality filter (`image_quality IN ('high', 'medium')`) for any training use.

7.3 Anthropological reference

The combination of `ethnicities` (structured group-level metadata) plus `notable_people` (Wikipedia-sourced examples) plus `image_observations` (per-image phenotype) supports anthropological survey work and educational use.

7.4 Multi-ethnic AI prompt grounding

Image-generation systems commonly default to phenotype-narrow outputs in the absence of explicit prompts. The synthesized `phenotype_profile` text on the `ethnicities` config provides editorial prose suitable for prompt-grounding multi-ethnic generation systems.

7.5 NOT for individual classification

We repeat the §6.3 caveat as a use restriction: this dataset describes ethnic groups in aggregate. It does not describe individuals as ethnic representatives. Individual-row classification, profiling, surveillance, or identification applications are explicitly out of scope.

---

8. Ethical Considerations

Source legality. All photographs are sourced from Wikipedia / Wikimedia Commons; per-image licenses vary (public domain, CC-BY-SA, various Creative Commons variants) but are uniformly permissive for derivative use with attribution. We redistribute *URLs*, not bytes; downstream consumers fetching the actual image data must comply with the per-image license at the source URL. Each row's `reference_url` field links to the originating Wikipedia article for license verification.

Aggregate vs individual. The `image_observations` rows are structured per-image, but the dataset's framing, aggregations (`image_observed_distribution`), and use restrictions (§7.5) are all aggregate-level. We considered whether to publish the per-image rows at all, given individual-identifiability concerns; we concluded that (a) the underlying photographs are already public on Wikipedia, (b) per-image rows are necessary for fairness-audit use cases that aggregate rows alone cannot support, and (c) explicit use restrictions in the dataset card constrain reuse direction.

Ethnic-group taxonomy. The 484-group taxonomy is a curatorial choice, not a natural kind. Some groups overlap (sub-groups of larger groups appear separately), some boundaries are contested, and the English-name canonicalization erases multilingual self-identifications. The taxonomy is a starting point for cross-group queries, not a definitive ethnographic ontology. Contributions and corrections are welcomed via the live catalog.

Adult-content brand. The publishing organization (`EthnicErotic`) operates an adult-creator platform alongside the catalog. The dataset itself is text-only and contains no adult content. We acknowledge that the brand may filter some downstream academic or institutional adoption; the dataset is published under CC BY 4.0 specifically to permit reuse and re-hosting under different attribution should adopters prefer.

9. Reproducibility

All construction code is open-sourced under Apache 2.0 at https://github.com/Agaveis/phenotype-catalog-pipeline. The repository contains the six pipeline stages, a Prisma schema excerpt for the three relevant database tables, the exact vision prompt with design notes, and a `.env.example` listing the required environment variables.

Stage	Script
Wikipedia people scrape	`scripts/scrape-wikipedia-notable-people.mjs`
Image URL discovery	`scripts/enrich-image-urls-from-wikipedia.mjs`
Vision-LLM analysis	`scripts/analyze-images-via-bedrock.mjs`
Per-group aggregation	`scripts/aggregate-image-observations.mjs`
HuggingFace export	`scripts/build-hf-dataset.mjs`
Phenotype-profile drafting (group-level)	`scripts/draft-phenotype-profiles-via-claude.mjs`

The Bedrock analysis script accepts `--concurrency` and `--limit` flags for batch testing. Authentication uses `fromIni()` from `@aws-sdk/credential-provider-ini` to avoid the default credential chain's stale-resolution failure mode on Windows hosts.

The dataset exports are deterministic: same input data + same scripts = same CSV/JSONL outputs (modulo timestamp fields). The release manifest at `huggingface-dataset/manifest.json` records the schema version (currently 2), generation timestamp, row counts, and per-config coverage statistics.

Replication pre-requisites:

- AWS credentials with Bedrock access in `us-east-1` (the Sonnet 4.6 inference profile is region-pinned).
- Anthropic Claude Sonnet 4.6 model access enabled in the AWS account.
- A SQL Server connection to the source `ethniclist` table, or a substitute group taxonomy.
- ~$45 USD in Bedrock vision inference budget for the full 5,668-image run.
- HuggingFace CLI authenticated to a writable dataset repository.

10. Future Work

Second-source diversification. The dominant bias is the Wikipedia source frame (§6.1). Two paths to dilute it:

1. **User-submitted images** under a community-contribution license. The live catalog at ethnicerotic.com plans to add a community-image submission flow; rows from that source would carry `source_type = 'community'` and broaden the gender / public-life / photographic-era distribution.
2. **A second public-domain corpus.** Adult-performer databases (e.g., IAFD) provide a different set of per-individual photographs with phenotype labels, biased differently from Wikipedia. Inclusion is being scoped; the source-attribution column already exists to support multiple sources cleanly.

Validation of model self-report. A subset of the `image_observations` rows could be re-annotated by human labelers and the calibration of `confidence` measured directly. We have not done this work and believe it would improve the dataset's utility for fairness-audit applications.

Methodology generalization. The pipeline is not specific to the ethnic-group taxonomy used here. The same Wikipedia-scrape → image-discovery → vision-LLM-annotation → SQL-aggregation pipeline can be adapted to any other group taxonomy with Wikipedia coverage (occupation, nationality, historical era, etc.). We invite replication.

References

Buolamwini, J., & Gebru, T. (2018). Gender Shades: Intersectional Accuracy Disparities in Commercial Gender Classification. *Proceedings of Machine Learning Research*, 81, 77–91. (Conference on Fairness, Accountability and Transparency.)

Fitzpatrick, T. B. (1988). The Validity and Practicality of Sun-Reactive Skin Types I Through VI. *Archives of Dermatology*, 124(6), 869–871.

Karkkainen, K., & Joo, J. (2021). FairFace: Face Attribute Dataset for Balanced Race, Gender, and Age for Bias Measurement and Mitigation. *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, 1548–1558.

Lhoest, Q., Villanova del Moral, A., Jernite, Y., et al. (2021). Datasets: A Community Library for Natural Language Processing. *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing: System Demonstrations*, 175–184.

Wang, M., Deng, W., et al. (2020). Mitigating Bias in Face Recognition Using Skewness-Aware Reinforcement Learning. (BUPT-Balancedface and RFW datasets.) *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*.
